2.5.4

Analogous to mean value theorem, show that for $n \geqslant 3, \Omega = B(0,r)$ and $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ such that

$$\begin{cases} -\Delta u = f, & x \in \Omega, \\ u = g, & x \in \partial\Omega \end{cases}$$

we have

$$u(0) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial\Omega} g(x)\,\mathrm{d}s(x) + \frac{1}{n(n-2)\alpha(n)} \int_{\Omega} (|x|^{2-n} - r^{2-n}) f(x)\,\mathrm{d}x.$$

Proof: Let $\phi(t) = \frac{1}{m(\partial B(0,t))} \int_{\partial B(0,t)} u(x)\,\mathrm{d}s(x)$ with $\phi(0) = u(0)$, then

$$u(0) = \phi(r) - \int_0^r \phi'(t)\,\mathrm{d}t$$

where $\phi(r) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial\Omega} g(x)\,\mathrm{d}s(x)$. Note that

$$\begin{aligned} \phi'(t) &= \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} \frac{d}{dt} u(tx)\,\mathrm{d}s(x) = \frac{1}{n\alpha(n)t^{n-1}} \int_{\partial B(0,t)} \nabla u(x) \cdot \vec{n}\,\mathrm{d}s(x) \\ &= \frac{1}{n\alpha(n)t^{n-1}} \int_{B(0,t)} \Delta u\,\mathrm{d}x = -\frac{1}{n\alpha(n)t^{n-1}} \int_{B(0,t)} f\,\mathrm{d}x. \end{aligned}$$

Hence

$$\begin{aligned} -\int_0^r \phi'(t)\,\mathrm{d}t &= \frac{1}{n\alpha(n)} \int_0^r t^{1-n} \int_{B(0,t)} f\,\mathrm{d}x\,\mathrm{d}t = \frac{1}{n\alpha(n)} \int_0^r t^{1-n} \int_0^t \int_{\partial B(0,s)} f(x)\,\mathrm{d}s(x)\,\mathrm{d}s\,\mathrm{d}t \\ &= \frac{1}{n\alpha(n)} \int_0^r \int_{\partial B(0,s)} f(x) \int_s^r t^{1-n}\,\mathrm{d}t\,\mathrm{d}x\,\mathrm{d}s \\ &= \frac{1}{n(n-2)\alpha(n)} \int_0^r (t^{2-n} - r^{2-n}) \int_{\partial B(0,t)} f(x)\,\mathrm{d}x\,\mathrm{d}s. \end{aligned}$$

2.5.5

If we let $n = 2$ in 2.5.4, show that

$$u(0) = \frac{1}{2\pi r} \int_{\partial\Omega} g\,\mathrm{d}l + \frac{1}{2\pi} \int_{\Omega} (\log r - \log|x|) f(x)\,\mathrm{d}x.$$

Proof: Analogous to 2.5.4, only that $\int_s^r t^{1-n}\,\mathrm{d}t = \log r - \log s$.

2.5.6

Suppose $v \in C^2(\Omega)$ satisfy $-\Delta v \leqslant 0$ on Ω, we call v subharmonic.
(1) Prove that for any $B(x,r) \subset \Omega$,

$$v(x) \leqslant \frac{1}{m(B(x,r))} \int_{B(x,r)} v(y)\,\mathrm{d}y. \tag{$\star$}$$

(2) Prove that $\max_{\overline{\Omega}} v = \max_{\partial\Omega} v$. (If Ω is bounded.)
(3) Suppose $\phi : \mathbb{R} \to \mathbb{R}$ is smooth and convex, u is harmonic on Ω, prove that $v = \phi(u)$ is subharmonic on Ω.
(4) Suppose u is harmonic on Ω, prove that $v = |Du|^2$ is subharmonic on Ω.
Proof: (a) Let $\phi(r) = \frac{1}{m(\partial B(x,r))} \int_{\partial B(x,r)} v(y)\,\mathrm{d}y = \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} v(rx)\,\mathrm{d}s(x)$, then

$$\phi'(r) = \frac{1}{n\alpha(n)} \int_{\partial B(0,r)} \nabla v(x) \cdot \vec{n}\,\mathrm{d}s(x) = \frac{1}{n\alpha(n)} \int_{B(0,r)} \Delta v\,\mathrm{d}x \geqslant 0,$$

hence $u(x) = \phi(0) \leqslant \phi(r)$, so

$$u(x) \leqslant \frac{1}{r} \int_0^r \phi(t)\,\mathrm{d}t = \frac{1}{m(B(x,r))} \int_{B(x,r)} v(y)\,\mathrm{d}y.$$

(b) Using (1) we can show that in each connected component of Ω, if an interior point x_0 reaches maximum, then v is constant on that component. Hence the global maximum must be reached at the boundary.
(3) By mean value theorem and Jensen's inequality, $v = \phi(u)$ satisfy $(\star)$. If $-\Delta v > 0$ in some open set $x \in U$, then $\phi'(r) < 0$ so $\phi(r) = \frac{1}{m(\partial B(x,r))} \int_{\partial B(x,r)} u(y)\,\mathrm{d}y$ is decreasing. Hence $v(x) > \frac{1}{m(B(x,r))} \int_{B(x,r)} v(y)\,\mathrm{d}y$ for $B(x,r) \subset U$, a contradiction.
(Or one can just use the chain rule.)
(4) Note that u_{x_i} are all harmonic so by (3) $u_{x_i}^2$ are subharmonic, hence their sum $|Du|^2$ is subharmonic.

2.5.7

Suppose $\{u_n\} \subset C(\overline{\Omega}) \cap C^2(\Omega)$ are harmonic on Ω, and $\{u_n\}$ converges uniformly on $\partial\Omega$. Show that $\{u_n\}$ converge uniformly on $\overline{\Omega}$ and the limit is harmonic.
Proof: Note that $u_n - u_m$ are harmonic, and its value on Ω is bounded by its value on $\partial\Omega$ by strong maximum principle, so for each $\varepsilon > 0$ take N such that $\forall n, m > N$, $\sup_{x\in\partial\Omega} |u_n(x) - u_m(x)| < \varepsilon$, then $\sup_{x\in\overline{\Omega}} |u_n(x) - u_m(x)| < \varepsilon$ so u_n converges uniformly on $\overline{\Omega}$. The limit u is still in $C(\overline{\Omega}) \cap C^2(\Omega)$, and by uniform convergence, u also satisfy mean value theorem, so u is harmonic.

2.5.10

Suppose $u(x)$ is harmonic on $\Omega = B(0,R)$ and continuous on $\overline{\Omega}$, and $M = \int_\Omega u^2\,\mathrm{d}x$. Prove that
(1) $|u(0)| \leqslant \sqrt{M/\alpha(n)R^n}$.
(2) $|u(x)| \leqslant \sqrt{M/\alpha(n)(R-|x|)^n}$.
Proof: (1) By Cauchy's inequality,

$$|u(0)|^2 = \frac{1}{\alpha(n)^2 R^{2n}} \left| \int_\Omega u \, \mathrm{d}x \right|^2 \leqslant \frac{1}{\alpha(n) R^n} \int_\Omega u^2 \, \mathrm{d}x = \frac{M}{\alpha(n) R^n}.$$

(2) Just replace Ω in (1) with $\Omega' = B(x, r)$ where $r = R - |x|$.

2.5.13

Suppose u is harmonic on $\mathbb{R}^n$ and $|u(x)| \leqslant C_1 |x|^m + C_2, x \in \mathbb{R}^n$, where $m \in \mathbb{N}, C_1, C_2 > 0$, then u is a harmonic polynomial of degree at most m.
Proof: We show that $\partial^\alpha u \equiv 0$ for each $|\alpha| = m + 1$, which implies u is a harmonic polynomial of degree at most m. By local estimates,

$$|\partial^\alpha u(x)| \leqslant \frac{C_{m+1}}{r^{n+m+1}} \int_{B(x,r)} |u(x)| \, \mathrm{d}x = O(r^{-1})$$

tends to zero as $r \to \infty$.

1.

Solve the following equation:

$$\begin{cases} u_t + b \cdot \nabla u + cu = 0, & \text{on } \mathbb{R}^n \times (0, \infty), \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

where $c \in \mathbb{R}, b \in \mathbb{R}^n$ are constant.
Solution: Let $z(s) = u(x + bs, t + s)$ then $z'(s) = b \nabla u + u_t = -cu = -cz(s)$, so $z(s) = z(-t) e^{-c(s+t)}$ $= g(x - tb) e^{-c(s+t)}$. Hence $u(x, t) = z(0) = g(x - tb) e^{-ct}$.